Customer Churn Prediction and Segmentation

An Analysis of Online Retail Transaction Data

Project Report | Data Science

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1. Introduction

For an online retailer, finding a new customer is usually more expensive than keeping an existing one. Most of the revenue comes from people who have already bought once and come back. The difficulty is that customers rarely announce they are leaving. They simply stop ordering, and by the time the business notices, they are gone.

This project works on that problem using the transaction history of a UK-based online gift retailer. It has two linked goals. The first is to group customers into meaningful segments based on how they buy. The second is to identify which customers are likely to churn, so that a business could act before they leave.

This report covers the work completed so far: understanding the dataset, cleaning it, and exploring it in enough depth to decide how segmentation and churn modelling should be done. The modelling stage is described in Section 8.

2. Problem Statement and Objectives

The dataset records what was bought and when, but it does not say which customers have left. There is no churn column, so churn has to be defined from purchasing behaviour before anything can be predicted.

The objectives of the project are:

1. Clean the raw transaction data so that it represents genuine customer purchases.
2. Study the data to understand sales patterns, product and country contribution, and customer behaviour.
3. Build customer-level Recency, Frequency and Monetary (RFM) measures.
4. Segment customers into groups with distinct behaviour.
5. Define churn in a defensible way and build a model that predicts it.

Objectives 1 to 3 are complete. Objectives 4 and 5 are the next phase.

3. Dataset

The data is the Online Retail dataset (UCI Machine Learning Repository). It contains every transaction of a registered UK online retailer between 1 December 2010 and 9 December 2011. The company mainly sells unique all-occasion gifts, and many of its customers are wholesalers. The raw file has 541,909 rows and 8 columns.

Column	Description
InvoiceNo	Invoice number. Invoices starting with "C" are cancellations
StockCode	Product code
Description	Product name
Quantity	Units bought in that line
InvoiceDate	Date and time of the transaction
UnitPrice	Price per unit in pounds sterling (£)
CustomerID	Unique customer identifier
Country	Country of the customer

One feature of this data matters for everything that follows: each row is one product line on an invoice, not one order. A single invoice has several rows, so counting rows and counting orders give very different answers.

4. Data Preprocessing

The raw data was copied into a working dataframe so the original stayed untouched. The InvoiceDate column was converted from text to a proper datetime type using the format month/day/year hour:minute. The following cleaning steps were then applied in order.

Step	Rows removed	Rows remaining
Raw data		541,909
Exact duplicate rows	5,268	536,641
Missing CustomerID	135,037	401,604

Step	Rows removed	Rows remaining
Cancelled invoices (InvoiceNo starts with "C")	8,872	392,732
Quantity or UnitPrice less than or equal to zero	40	392,692

The reasoning behind each step:

- Duplicates were removed because a repeated identical line is almost certainly a recording error and would inflate quantities.
- Rows without a CustomerID were dropped. Both segmentation and churn are defined per customer, so a transaction that cannot be linked to a customer has no use here. This was the largest removal, about a quarter of the raw data.
- Cancelled invoices were removed so that the analysis reflects purchases rather than returns.
- Non-positive quantities and prices were removed as invalid. Only 40 rows were affected after the earlier steps.

A new column, TotalAmount = Quantity × UnitPrice, was added to measure the value of each line. After cleaning, 392,692 rows (72.5% of the raw data) remain. No missing values are left in the Description column.

5. Exploratory Data Analysis

5.1 The data at a glance

Measure	Value
Transactions (rows)	392,692
Unique customers	4,338
Unique invoices	18,532
Unique stock codes	3,665
Countries	37
Period covered	1 Dec 2010 to 9 Dec 2011
Total revenue	£8,887,209

The typical line is small. The median Quantity is 6 units, the median UnitPrice is £1.95 and the median TotalAmount is £12.45. The averages are much higher (13 units, £3.13 and £22.63), and the maximum Quantity is 80,995. A gap this wide between median and mean

is the first sign that the data is heavily skewed. At invoice level, the median order is worth £302.58.

5.2 Outliers

Outliers were checked with boxplots and the 1.5 × IQR rule.

Variable	Upper bound	Outlier rows	Share of rows	Share of revenue from high outliers
Quantity	27	25,616	6.52%	40.06%
UnitPrice	7.50	34,112	8.69%	13.98%
TotalAmount	42.07	31,231	7.95%	50.79%

The last column is the important one. Roughly 8% of lines are flagged as outliers on TotalAmount, yet they carry just over half of all revenue. These are largely bulk wholesale orders rather than mistakes, so deleting them by rule would throw away the most valuable part of the business. The decision taken is to keep them and handle the skew with a log transformation instead (see 5.3). The few cases that are real errors are discussed in Section 7.

5.3 Distributions and the effect of a log transform

Quantity, UnitPrice and TotalAmount are all strongly right-skewed: most values are small and a thin tail of very large values stretches the scale. After applying log(1 + x), the distributions are far more symmetric. This matters later because K-Means clustering uses distances and would otherwise be dominated by a handful of extreme customers.

5.4 Products

The five best-selling products by units are Paper Craft Little Birdie (80,995), Medium Ceramic Top Storage Jar (77,916), World War 2 Gliders Assorted Designs (54,319), Jumbo Bag Red Retrospot (46,078) and White Hanging Heart T-Light Holder (36,706).

Ranked by revenue the picture changes. The leader is still Paper Craft Little Birdie (£168,470), but second place goes to the Regency Cakestand 3 Tier (£142,265), followed by the White Hanging Heart T-Light Holder (£100,392). A product that sells many units at a low price and a product that sells fewer units at a higher price do not rank the same way, which is why both views are needed.

Two entries in the revenue top ten are not products at all: POST (postage, £77,804) and M (manual adjustments, £53,420). This is addressed in Section 7.

5.5 Geography

The business is heavily concentrated in one market. The United Kingdom accounts for 3,920 of the 4,338 customers (90%), 16,646 of the 18,532 invoices, and £7.29 million of revenue, which is 82% of the total.

The next largest markets by revenue are the Netherlands (£285,446), EIRE (£265,262), Germany (£228,678), France (£208,934) and Australia (£138,454). Some of these have very few customers. The Netherlands has only 9 customers yet ranks second, and EIRE has 3. In these countries a single large buyer can account for most of the revenue, so country-level figures should be read with care.

5.6 Revenue over time

Monthly revenue stays between about £450,000 and £680,000 from December 2010 to August 2011, then rises sharply: £950,690 in September, £1,035,642 in October and £1,156,206 in November. These three months alone account for about 35% of all revenue in the dataset. November is the peak at 13% of total revenue, which fits a business selling gifts ahead of Christmas.

The same pattern appears in customer counts. Active customers per month grow from roughly 750 to 1,000 through the early part of the year, to 1,664 in November. The December 2011 figure looks like a collapse (£517,190) but it is not one, because the data ends on 9 December and the month is only partly covered. It must not be read as customers leaving.

5.7 Customer behaviour (RFM)

Three measures were calculated for each of the 4,338 customers.

- Recency: days since the customer's last purchase, measured from one day after the final date in the data.
- Frequency: number of distinct invoices.
- Monetary: total amount spent.

	Mean	Median	Maximum
Recency (days)	92.5	51	374
Frequency (invoices)	4.27	2	209
Monetary (£)	2,049	669	280,206

Again the mean is well above the median for Frequency and Monetary, so a small group of customers behaves very differently from the typical one. The skewness values confirm this: 12.1 for Frequency and 19.3 for Monetary before transformation, falling to 1.2 and 0.4 after log(1 + x). Recency is much milder (1.2 before, -0.4 after).

Correlations between the log-transformed measures:

	Recency	Frequency	Monetary
Recency	1.00	-0.57	-0.48
Frequency	-0.57	1.00	0.81
Monetary	-0.48	0.81	1.00

Frequency and Monetary move together strongly (0.81): customers who order often also spend more in total. Recency is negatively related to both, meaning customers who bought recently tend to buy more often and spend more. This is the relationship churn modelling relies on, because a customer drifting away shows up first as rising Recency. The high Frequency-Monetary correlation also means these two carry overlapping information, which is worth remembering when interpreting clusters.

Two further measures describe how loyal the customer base is.

- 1,493 customers (34.4%) bought only once. The remaining 2,845 (65.6%) are repeat buyers. More than a third of customers never came back after their first order, which is the clearest sign that churn is a real issue for this business.
- The top 20% of customers (867 people) generate 74.7% of revenue. This is close to the classic 80/20 pattern and shows how much depends on a small group of valuable customers.

6. Key Findings

1. After cleaning, 72.5% of the raw data is usable. A quarter of the original records had no customer ID and could not be used for customer-level analysis.
2. All the transaction variables are strongly right-skewed. A log transformation is necessary before clustering.
3. Bulk orders are a real part of the business: lines flagged as outliers are about 8% of rows but about half of revenue, so they are retained, not deleted.
4. Revenue is highly seasonal, with September to November producing about 35% of the total.
5. The UK dominates, with 82% of revenue and 90% of customers.

6. 34.4% of customers purchased only once.
7. 20% of customers produce about 75% of revenue.
8. Recency is clearly linked to Frequency and Monetary value, which supports using RFM features for both segmentation and churn prediction.

7. Limitations and Issues Identified

While examining the largest values, several problems were found that affect how reliable the numbers above are. They are recorded here with the planned correction for each.

Reversed orders remain in the data. The two biggest sales lines are Paper Craft Little Birdie (80,995 units, £168,470, customer 16446, 9 December 2011) and Medium Ceramic Top Storage Jar (74,215 units, £77,184, customer 12346, 18 January 2011). Checking the raw file shows that each was cancelled within minutes, by invoices C581484 and C541433. Removing only the cancellation rows, as was done in preprocessing, deletes the reversal but keeps the original sale. These two lines alone represent about 2.8% of recorded revenue that never actually happened. They also distort customer figures: customer 16446 appears among the top five spenders with £168,473 from just two invoices. The correction is to match cancellations to the invoices they reverse and drop both sides, or alternatively to net the cancellation quantities against the original sales.

Non-product codes. Rows with the stock codes POST (1,099 rows), M (279), C2 (133), DOT (16), BANK CHARGES (12) and PADS (3) are fees and adjustments rather than products. They total 1,542 rows and about 1.7% of revenue, but they appear in the product rankings and the highest unit prices (a single POST line of £8,142.75). They will be filtered out before product-level analysis and feature creation.

Cancellations carry information. Cancelled invoices were removed completely. A customer's return behaviour may itself predict churn, so a return-rate feature could be built from the cancellation data before it is discarded.

Partial final month. The data ends on 9 December 2011, so December cannot be compared fairly with other months.

One year of data. With a single Christmas period, seasonal effects cannot be separated from real changes in customer behaviour. This makes the choice of churn window more sensitive.

Recency uses the whole timeline. Recency here was measured from the last date in the full dataset. That is fine for describing customers, but for churn prediction, features must be computed only from the period before the prediction window. Otherwise the model would be given information from the future. Recency will be recalculated inside the observation window.

8. Methodology for the Remaining Work

Defining churn. The timeline will be split into an observation period, used to build features, and a prediction period that follows it. A customer who purchased during the observation period but made no purchase in the prediction period is labelled as churned. The length of the window will be chosen from the data by looking at the typical gap between a customer's consecutive purchases, so that the cut-off is supported by evidence and not picked arbitrarily. The share of churned customers will be checked, since an imbalanced target affects how the model should be evaluated.

Segmentation. Recency, Frequency and Monetary will be log-transformed and standardised, then clustered with K-Means. The number of clusters will be selected using the elbow method together with the silhouette score. Each cluster will be profiled by its average RFM values and given a business label, for example loyal high-value customers, new customers, at-risk customers and lost customers.

Churn modelling. A logistic regression will serve as the baseline, compared with tree-based models such as random forest and gradient boosting. Additional features will include average order value, number of distinct products, purchase gaps and return rate. Because churners may be a minority, performance will be judged with precision, recall, F1-score and ROC-AUC rather than accuracy alone, using a stratified train-test split.

Using the results. The final step links the two parts: which segments have the highest predicted churn, and what retention action suits each one.

9. Conclusion

The work so far turned a raw transaction log of 541,909 rows into a clean customer-level dataset of 4,338 customers with Recency, Frequency and Monetary measures. The analysis showed a business that is seasonal, concentrated in the UK, and dependent on a small group of high-spending customers, while more than a third of its customers never place a second order. The strong link between recency, frequency and spend gives good grounds for using these features to segment customers and predict churn.

The exploration also exposed data problems, mainly reversed orders that survived cleaning and non-product codes, and these have clear fixes that will be applied before modelling. The next phase builds the churn label, creates the segments and trains the prediction models on top of this groundwork.

References

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